

Bryan Zeng

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EXPERIENCE

INTEGRITY Security Services

Boston, MA

Full-Stack Software Engineer

January 2024 – Present

- Architected and deployed full-stack web applications that support the **provisioning of thousands of IEEE 1609.2 certificates** used in V2X processes, leveraging extensive use of AWS technology for rapid development and security
- Developed Kotlin Android and Swift iOS apps for workzone security, enabling **real-time interoperability** between roadside units, vehicles, and mobile devices over **MQTT, PC-5, and UDP** and improving on-site hazard awareness
- Implemented Kotlin and Swift signing libraries **using ECDH key exchange** for secure, signed V2X communication
- Optimized system observability by **migrating 32M+ logs** to DocumentDB, accelerating queries and cutting overhead
- Containerized EV2G app and ensured **reliable CI/CD** by creating Jenkins pipelines for build, deployment, and testing
- Built Flask microservice for **traffic intersection validation**, enabling Python-based validation from the web app
- Consolidated three repos into a monorepo to **facilitate end-to-end testing** and increase reusability and extensibility

Wolters Kluwer - UpToDate

Waltham, MA

Product Full-Stack Software Engineer

January 2023 – December 2023

- Planned and led initiatives using Spring Boot, REST APIs, and Vue.js to **eliminate technical debt**, resolve security vulnerabilities, and modernize legacy code for a medical resources platform **serving 2M+ health professionals**
- Enhanced internal developer tooling **used by 150+ developers** to ensure proper backend configuration
- Revamped front-end architecture by unifying error-handling templates, enhancing user experience and maintainability

NURover

Boston, MA

Software Developer | Webmaster

September 2021 – May 2025

- Co-developed a **small-scale haptic arm** in C++ as a POC of controlling a full 4-degree-of-freedom robotic arm
- Engineered **Python ROS state machine**, enabling autonomous Rover navigation via Zed2 camera GPS and IMU data

PROJECTS

Cheffy: AI Cooking Assistant

September 2025 – Present

- Co-architected Cheffy, an **agentic search AI** combining **Gemini, RAG, external API tooling**, and **custom recipe data** to autonomously retrieve recipes and provide personalized step-by-step cooking guidance

Finance Tracker AI Chatbot

June 2025 – Present

- Developed a **RAG AI chatbot using LangChain–Ollama** to provide natural language insights to transactions and budgets, automating financial analysis while supporting smarter spending habits

Twitter Sentiment Analysis Stock Portfolio

October 2022 – December 2022

- Built a High-Minus-Low trading model using NLTK, and Scikit-Learn from **sentiment across 1.75M+ tweets**

YouTube Comment Spam Classifier

February 2023 – May 2023

- Engineered a Tensorflow/Keras **deep learning classifier**, achieving **90.6% accuracy** on 2,000+ YouTube comments

EDUCATION

Northeastern University - Khoury College of Computer Sciences

Boston, MA

M.S. Candidate in Computer Science

Expected May 2026

B.S. in Computer Science and Business Administration, Concentration in Business Analytics

- **Honors: GPA: 3.985 / 4.0**, Dean's List
- **Relevant Coursework:** Foundations of Artificial Intelligence, Machine Learning I, Software Engineering, Networks, Database Management, Systems, Algorithms and Data, Object-Oriented Design, Foundations of Data Science

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Java, Python, Go, Kotlin, Swift, SQL, C, R, C++
- **Frameworks:** React, Node, Vue, Spring, Flask, Pandas, Ollama, LangChain, Scikit-Learn, TensorFlow, Streamlit
- **Technologies:** AWS, Docker, Git, Jira, Confluence, Jenkins, MongoDB, SQL Developer, Tableau

INTERESTS

Matcha, Cooking, Volleyball, Tennis, Machine Learning and AI